

# Arterial Blood Pressure Regulation

NEW YORK INSTITUTE  
OF TECHNOLOGY

College of Osteopathic  
Medicine

Principles and Practices of Osteopathic Medicine II



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Do.  
Make.  
Heal.  
Innovate.  
Reinvent the Future.

# Learning Objectives

1. Explain and summarize the mechanisms underlying **short-term** blood pressure regulation by baroreceptors, chemoreceptors, atrial reflexes, and the central nervous system-ischemic response.
2. Explain the mechanism underlying **long-term** blood pressure regulation via baroreceptors in juxtaglomerular apparatus, including step-by-step activation of renin-angiotensin-aldosterone system.
3. Interpret the integrated responses compensating for a drop in arterial blood pressure as a result of hypovolemic shock caused by acute bleeding.
4. Discuss clinical applications of blood pressure regulation concepts

# Review of Definitions

**Stroke Volume (SV):** Volume of blood pumped by the L/R ventricle in a single heartbeat. Affected by contractility, afterload, and preload.

**Cardiac Output (CO):** Volume of blood the heart pumps through the circulatory system **per minute**. CO ~5 L/min.

$$CO = SV \times HR$$

**Total Peripheral Resistance (TPR):** Amount of **resistance to blood flow** in the systemic circulation. Also referred to as systemic vascular resistance (**SVR**).

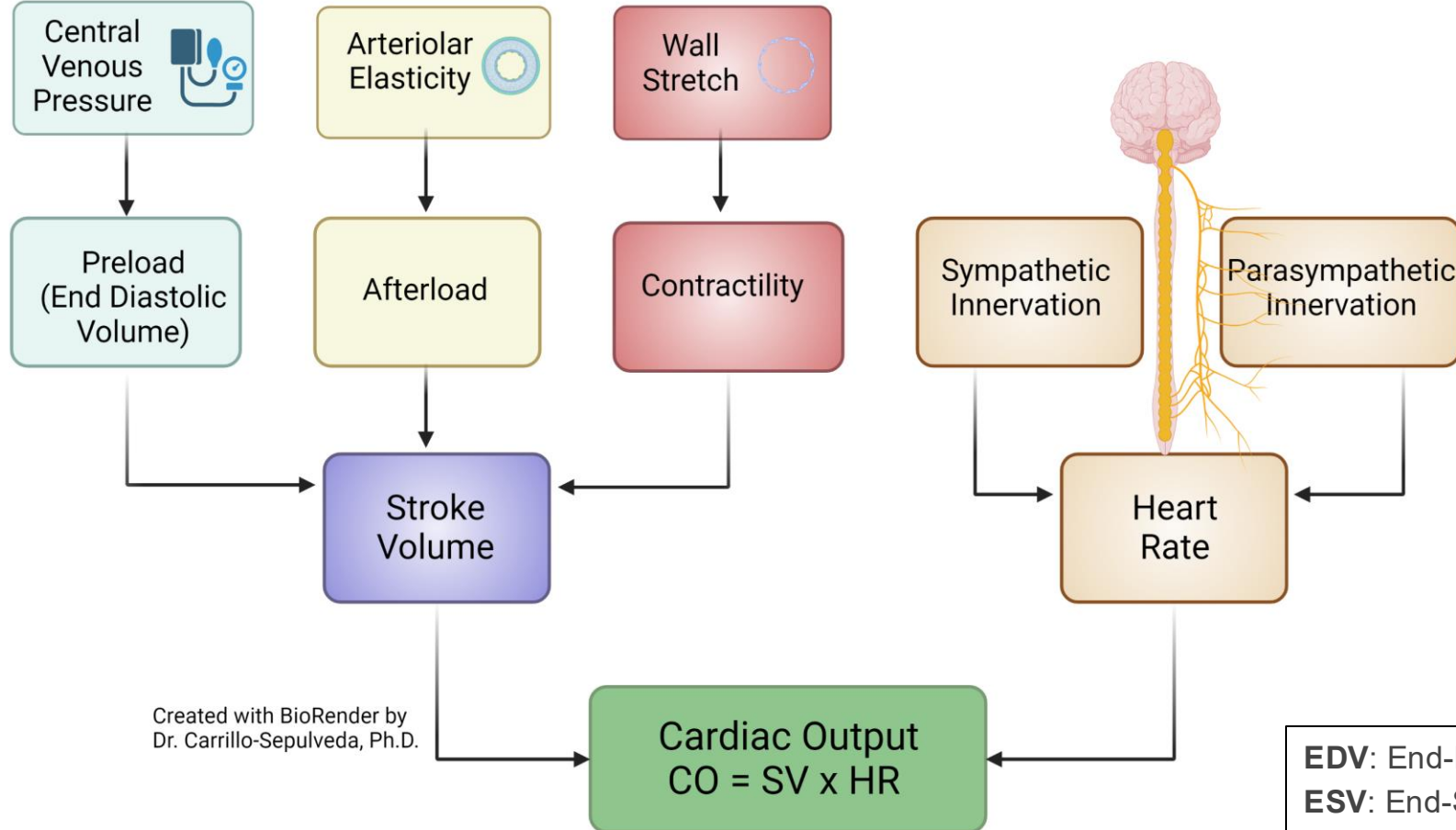
$$TPR = MAP / CO$$

**Mean Arterial Pressure (MAP):** The **average arterial pressure** throughout **one cardiac cycle**. MAP  $\geq$  60 mmHg to maintain adequate tissue perfusion.

$$MAP = CO \times TPR$$

$$MAP = 2 \times (\text{diastolic}) + \text{systolic} / 3$$

# Relationship of Cardiac Output, Stroke Volume and Heart Rate

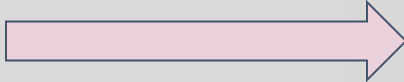


**EDV:** End-Diastolic Volume  
**ESV:** End-Systolic Volume

# BLOOD PRESSURE - HYDRAULIC EQUATION

## Hydraulic Equation:

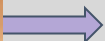
$$\text{Arterial Blood Pressure} = \text{CO} \times \text{TPR}$$

↑ CO or ↑ TPR  ↑ BP

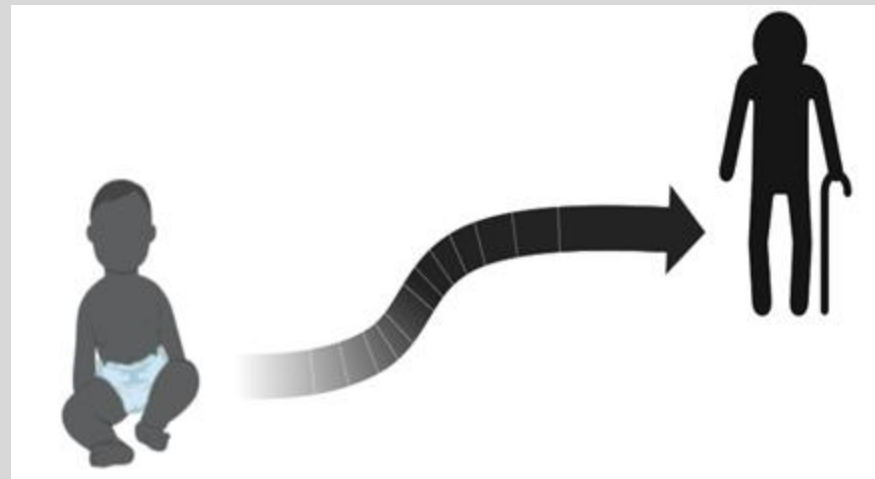
↓ CO or ↓ TPR  ↓ BP

# Blood Pressure Regulation

## Short-Term BP Regulation



## Long-Term BP Regulation



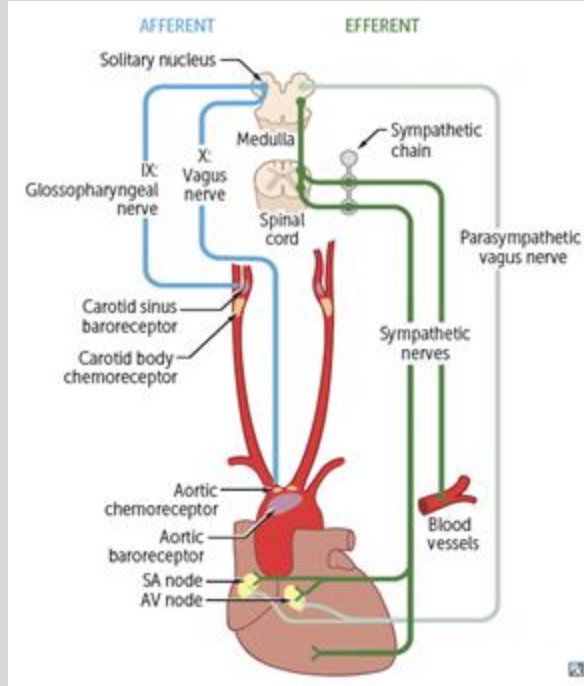
- BP is the driving force of blood flow to the organs in the body:
  - Thus, there is a complex set of systems in order to maintain BP at a constant level despite changes in the environment, short-term & long-term.

Image from: Biorix

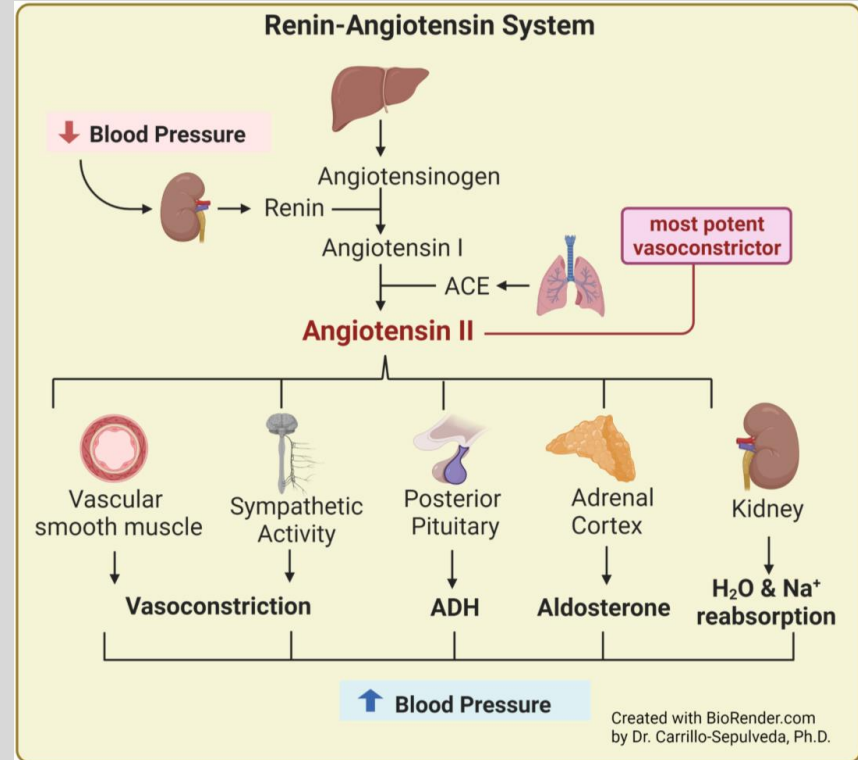
# Blood Pressure Regulation

Two physiologic systems primarily responsible for BP management

## Short-Term BP Regulation



## Long-Term BP Regulation



# Short-Term Blood Pressure Regulation

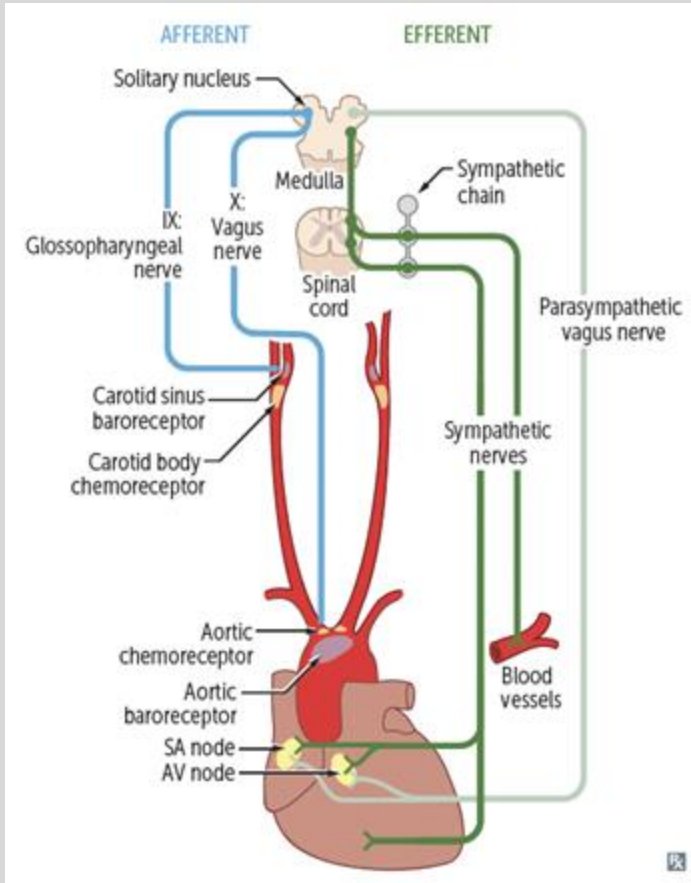
- ♥ Regulate blood vessel diameter, HR, contractility
- ♥ Short-term: Within seconds to minutes
- ♥ Occur through neural pathways

## **Key Players:**

- Baroreceptor Reflex
- Chemoreceptor Reflex
- CNS-Ischemic Response
- Atrial Reflex



# Short-Term BP Regulation



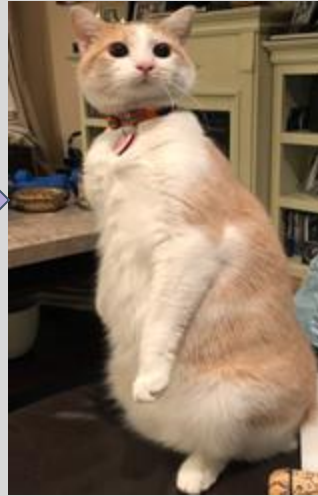
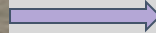
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DETECTORS</b>      | <p><b>PRIMARY SENSORS: BARORECEPTORS</b></p> <ul style="list-style-type: none"> <li>• Detect stretch of vascular wall</li> <li>• Located in aortic arch &amp; carotid sinus</li> </ul> <p><b>SECONDARY SENSORS: CHEMORECEPTORS</b></p> <ul style="list-style-type: none"> <li>• Detect changes in blood PO<sub>2</sub>, PCO<sub>2</sub>, and pH</li> <li>• Located in aortic arch &amp; carotid sinus</li> </ul> |
| <b>AFFERENT</b>       | <p><b>CN IX (GLOSSOPHARYNGEAL NERVE)</b></p> <ul style="list-style-type: none"> <li>• Pathway for impulses</li> <li>• Located in carotid sinus</li> </ul>                                                                                                                                                                                                                                                        |
| <b>AFFERENT</b>       | <p><b>CN X (VAGUS NERVE)</b></p> <ul style="list-style-type: none"> <li>• Pathway for impulses</li> <li>• Located in aortic arch</li> </ul>                                                                                                                                                                                                                                                                      |
| <b>CONTROL CENTER</b> | <p><b>NUCLEUS TRACTUS SOLITARIUS (NTS)</b></p> <ul style="list-style-type: none"> <li>• Receives impulses and responds</li> <li>• Located in medulla</li> </ul>                                                                                                                                                                                                                                                  |
| <b>EFFERENT</b>       | Pathway for CNS response to BP changes                                                                                                                                                                                                                                                                                                                                                                           |
| <b>EFFECTORS</b>      | <ul style="list-style-type: none"> <li>• Pacemaker and muscle cells in the heart</li> <li>• Vascular smooth muscle cells in arteries and veins</li> <li>• Adrenal medulla</li> </ul>                                                                                                                                                                                                                             |

# Baroreceptor Reflex

Marichelle Pitta, OMS Scholar

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When you realize you are late and jump up quickly from your bed...



Positional change



**Baroreceptor  
reflex  
activated**



Increased  
sympathetic  
activity



Effects

↑HR, ↑SV, ↑BP

PALPITATIONS

DIZZINESS

# Baroreceptors

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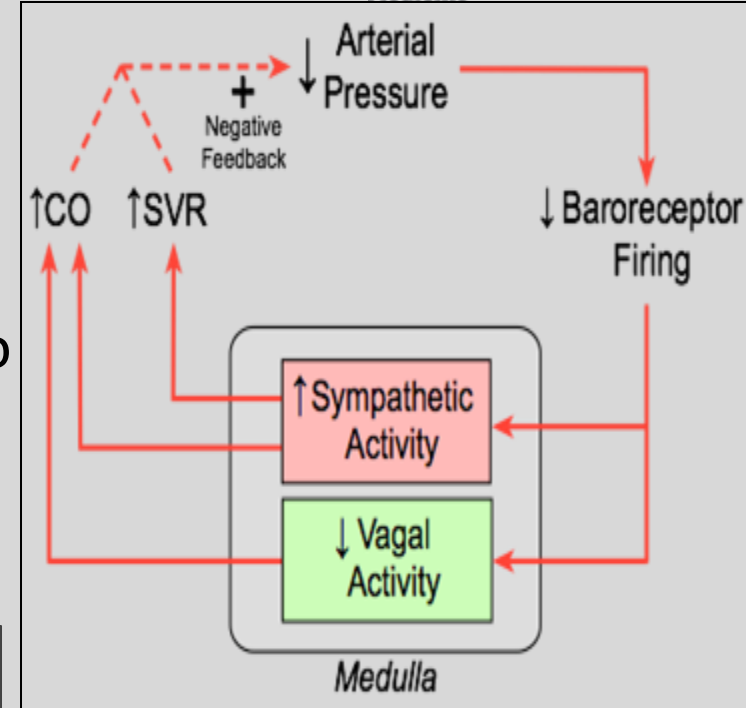
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- Stretch-sensitive
- Located in **carotid sinuses** and **aortic arch**
- **Sense changes in BP**
- **Send *afferent* signals** to brain stem (NTS)

via

**CN IX**  
(Carotid  
sinus)

**CN X**  
(Aortic  
arch)



NTS: Nucleus Tractus Solitarius

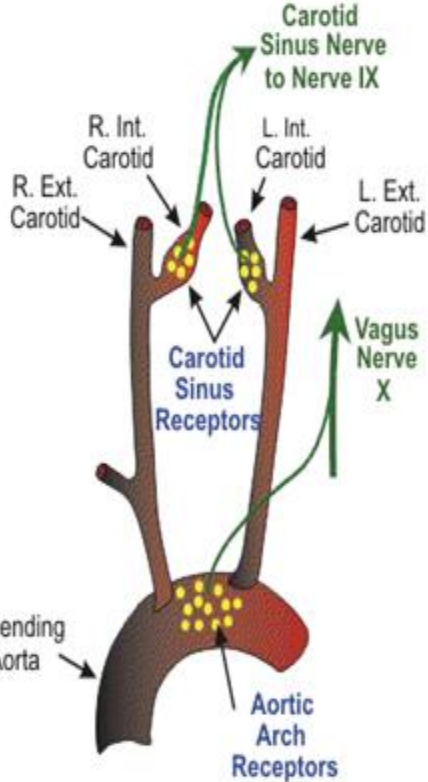
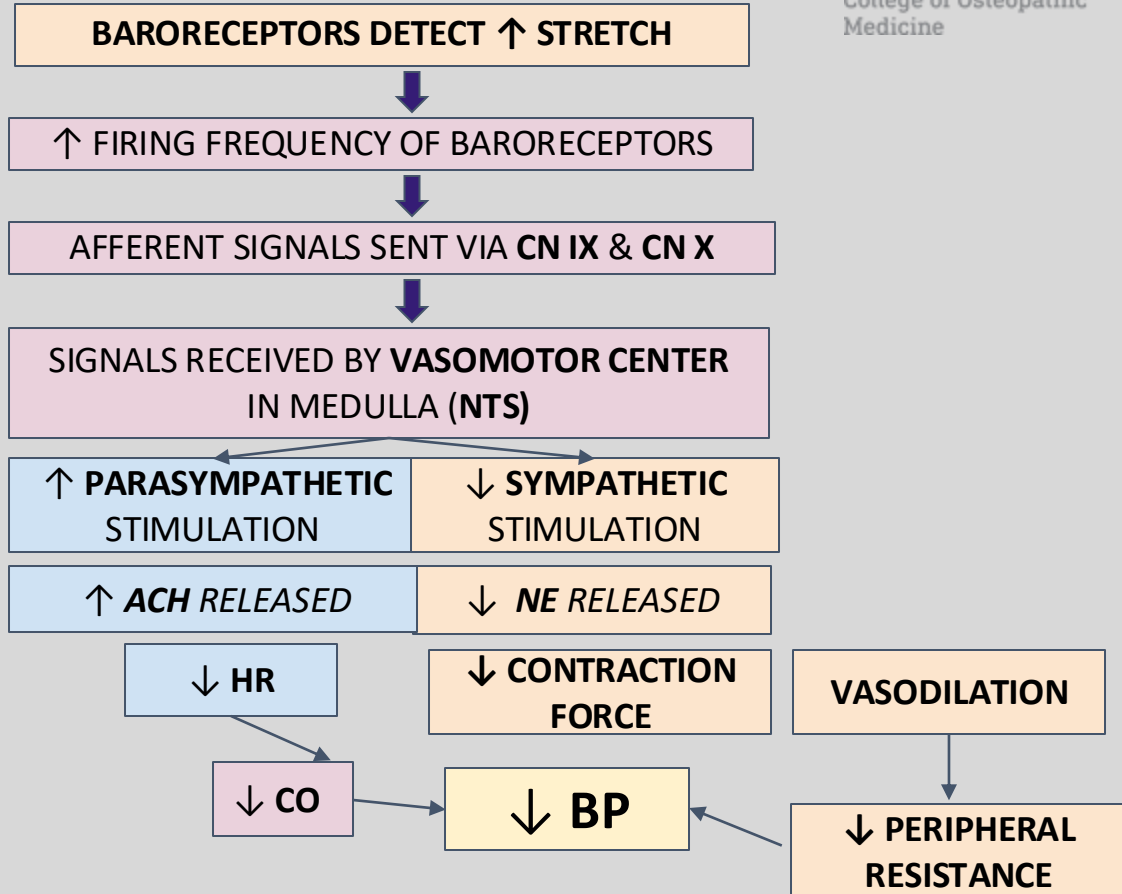


Figure 1. Location and innervation of arterial baroreceptors.

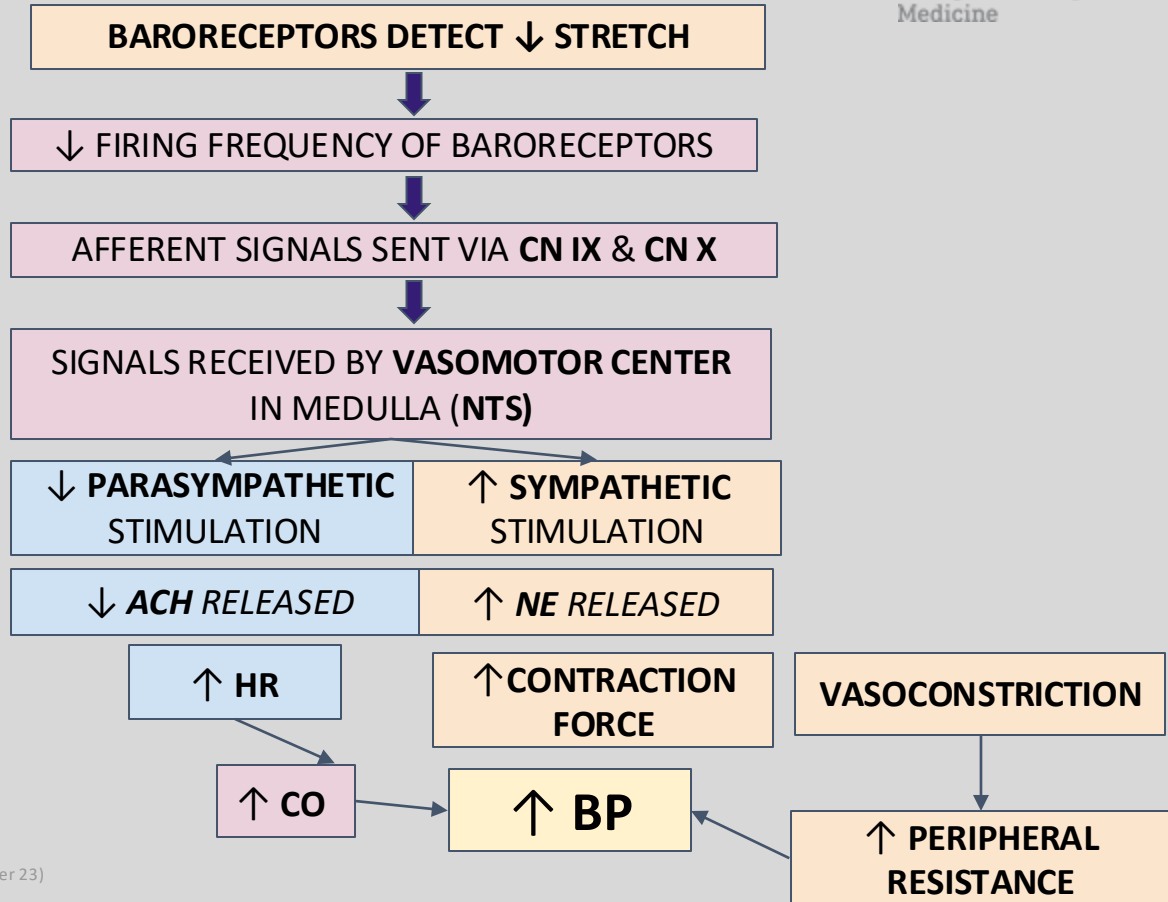
# BARORECEPTOR Reflex - Mechanism

↑ INCREASED BP



# Baroreceptor Reflex - Mechanism

↓ **DECREASED BP** 



# Baroreceptor - VALSALVA MANEUVER

- ♥ FORCEFUL EXHALATION AGAINST A CLOSED AIRWAY
- ♥ CAN BE USED TO ↓ HR (EXAMPLE: Supra Ventricular Tachicardia))

↑ INTRATHORACIC PRESSURE



↑ BARORECEPTOR STRETCH



↑ BARORECEPTOR FIRING

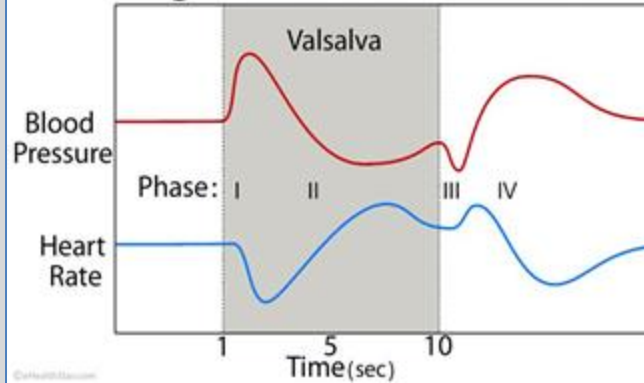


↑ PARASYMPATHETIC &  
↓ SYMPATHETIC ACTIVITY



↓ HR

## Blood Pressure and Heart Rate During the Valsalva Maneuver



## Valsalva maneuver



## VALSALVA MANEUVER - PHASES:

PHASE 1 (START STRAIN)

PHASE 2 (CONTINUED STRAIN)

PHASE 3 (RELEASE OF STRAIN)

PHASE 4 (RECOVERY)

↑ Intrathoracic  
Pressure

↓ Venous return  
(preload)

↓ CO

↓ Intrathoracic  
Pressure

↓ Afterload

↑ SV

↑ CO

# Baroreceptors: Carotid Massage & Carotid Sinus Syndrome

## CAROTID MASSAGE

- ♥ Stimulates the baroreceptors located in the carotid sinus
- ♥ Can be used to ↓HR



### CONTRAINDICATION

–Avoid carotid massage in patients with **prior TIA/CVA** or those with **carotid bruits**

↑Risk of breaking small emboli during carotid massage if patient has **carotid artery stenosis**, can lead to TIA/CVA

CAROTID MASSAGE SIMULATES ↑BP



↑PRESSURE ON CAROTID SINUS  
BARORECEPTORS



↑BARORECEPTOR FIRING



CN IX TRANSMITS NERVE IMPULSE TO  
MEDULLA (NTS)



NTS RESPONDS VIA CN X (↑  
PARASYMPATHETIC ACTIVITY)



↑AV NODE REFRACTORY PERIOD



↓HR

## CAROTID SINUS HYPERSENSITIVITY SYNDROME

↑ Carotid sinus sensitivity

- Frequently a/w arteriosclerotic changes in carotid sinus (Common in older patients)

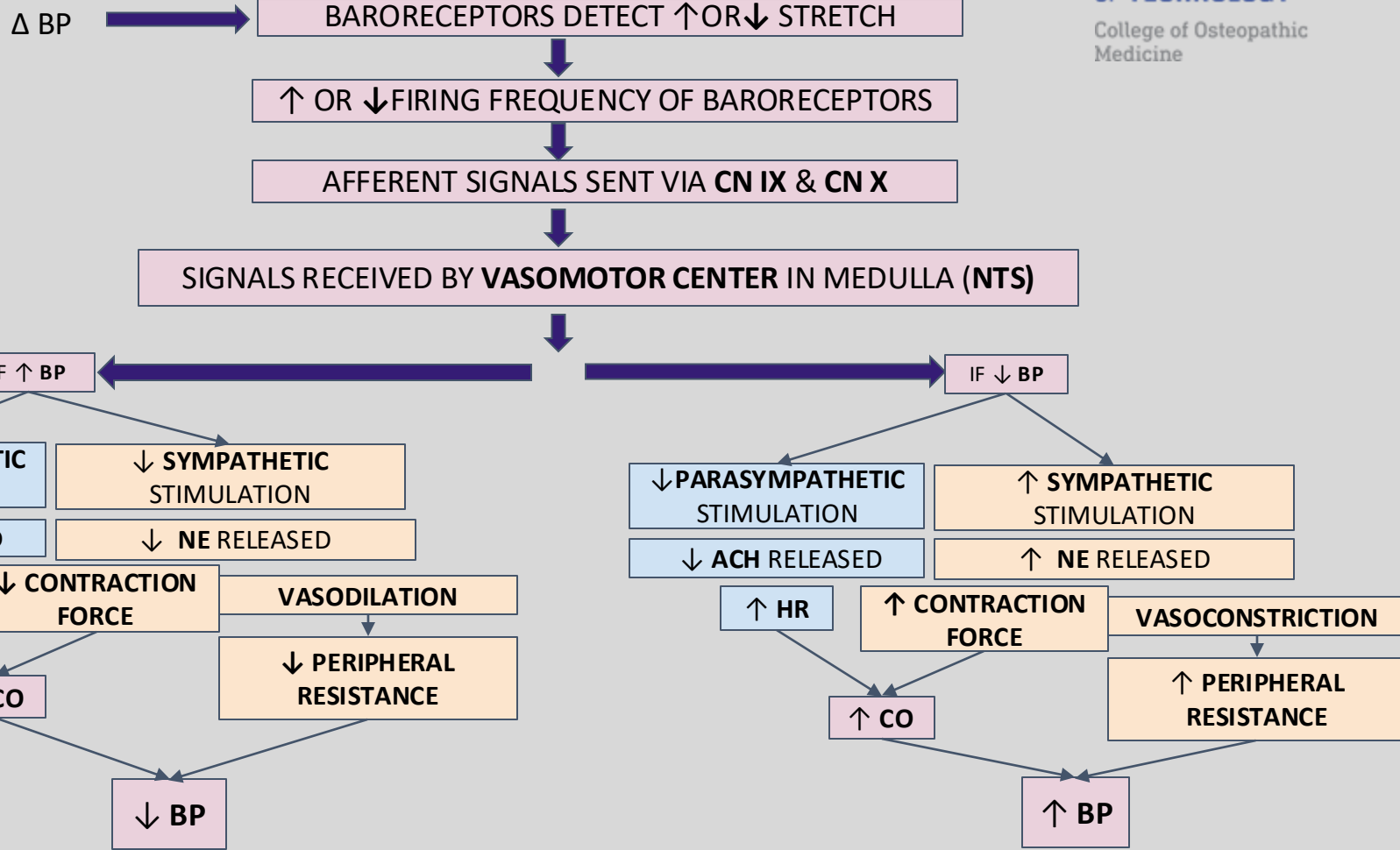
### Risk factors:

- Older age
- Male
- Atherosclerotic disease
- Prior neck surgery/neck irradiation

### Board exam clinical vignette:

*Elderly male patient presenting with syncope/pre-syncopal episode, occurred while buttoning shirt or shaving.*

# SUMMARY - SHORT-TERM BP REGULATION - BARORECEPTORS





# Chemoreceptors

- ♥ Specialized receptors, secondary sensors.
- ♥ Detect changes in blood  $O_2$ ,  $CO_2$ , and pH.
- ♥ Types: **Peripheral** and **Central**

## Mechanism

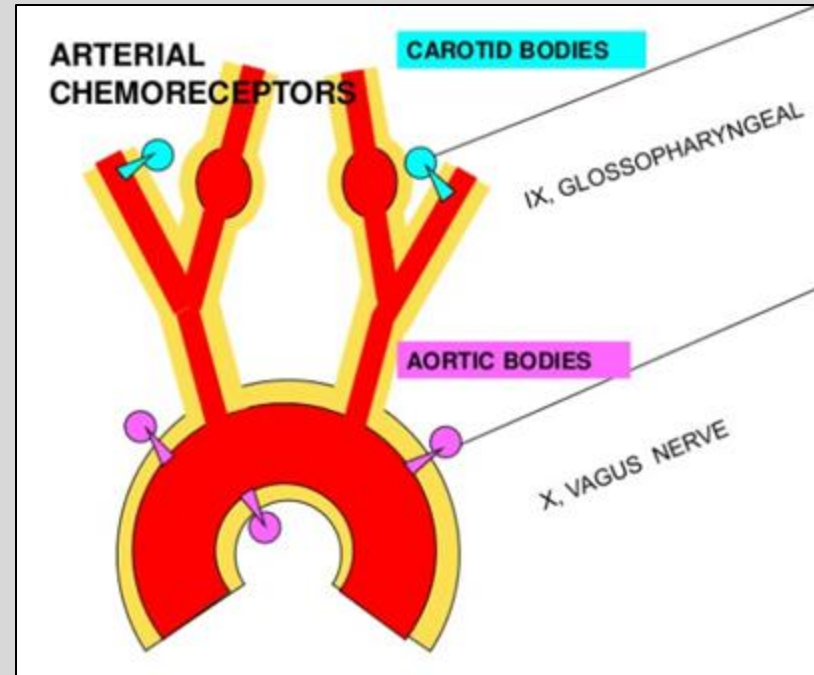
**Peripheral chemoreceptors** (in carotid & aortic bodies) detect  $\uparrow CO_2$ ,  $\downarrow O_2$ ,  $\downarrow pH$

Afferent signals sent via **CN IX & CN X**

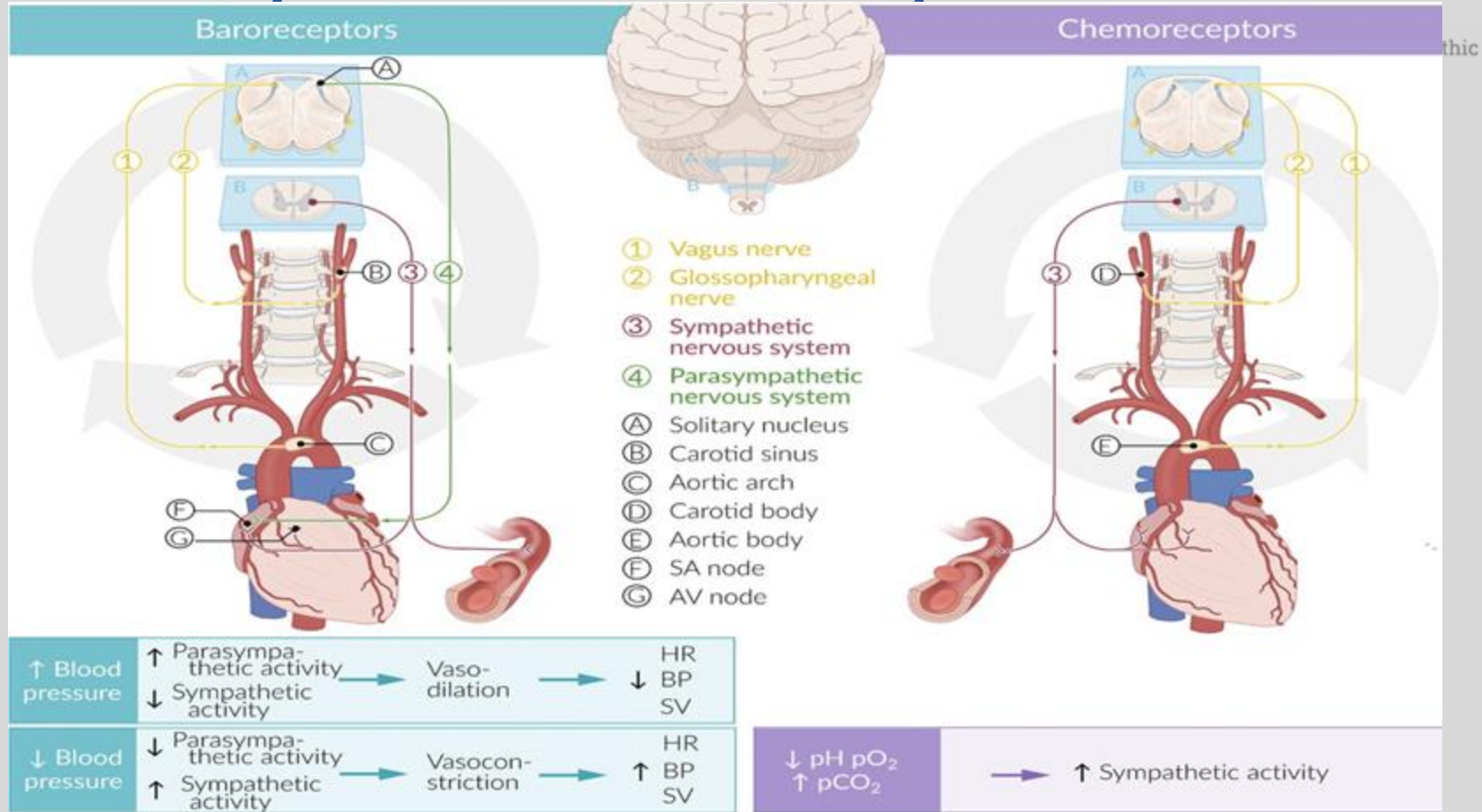
Stimulation of medullary respiratory center & medullary VMC

$\uparrow$  Sympathetic activity

$\uparrow BP$

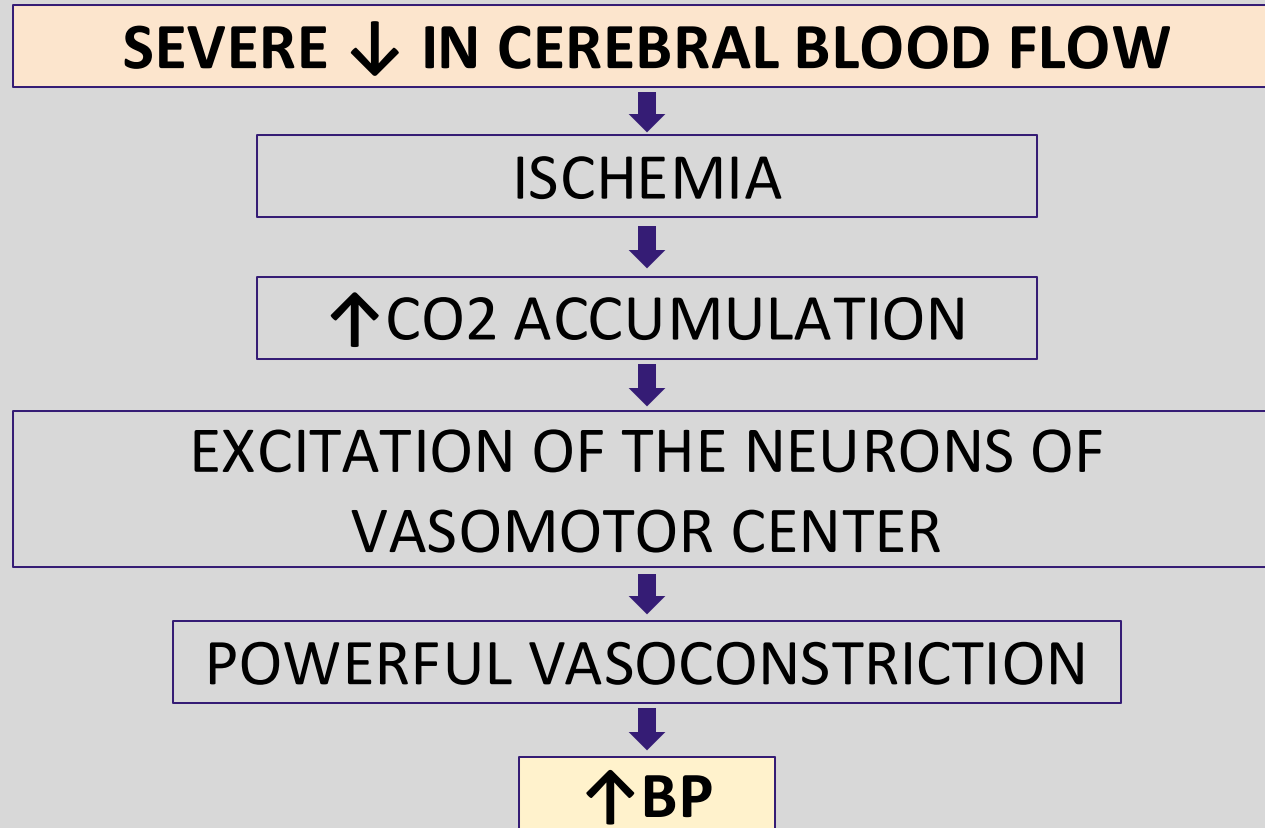


# Baroreceptor and Chemoreceptor Reflex



# CNS-ISCHEMIC REFLEX

Occurs when there is a severe decrease in blood flow to the brain



# CUSHING REFLEX

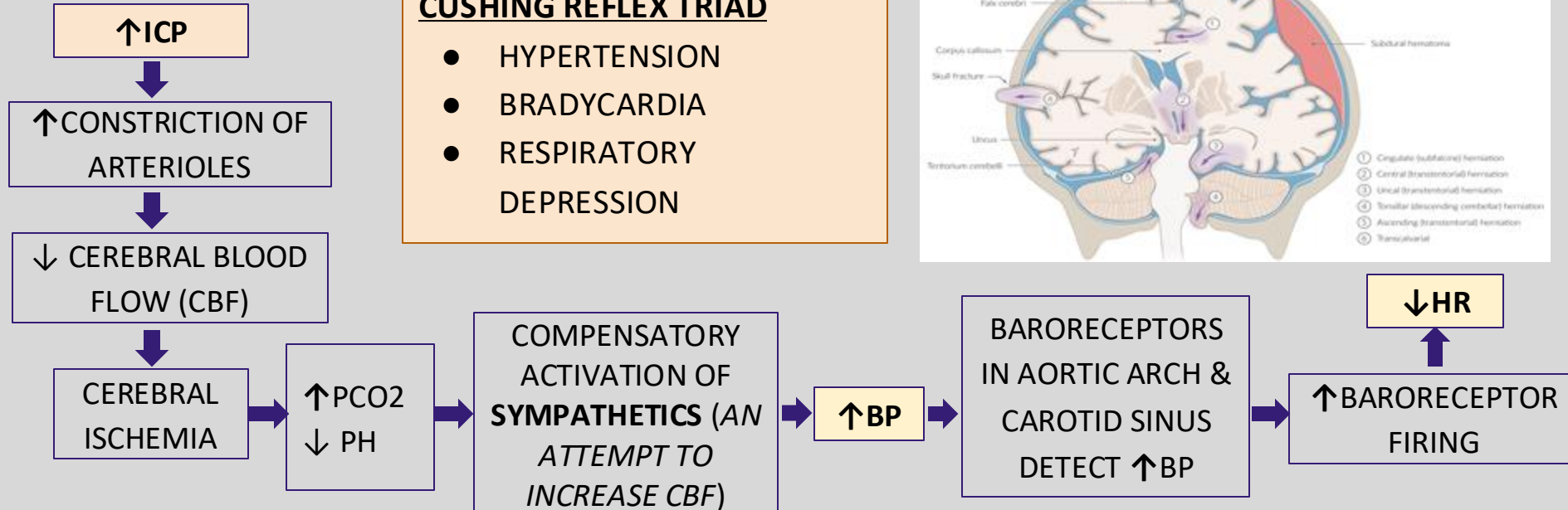
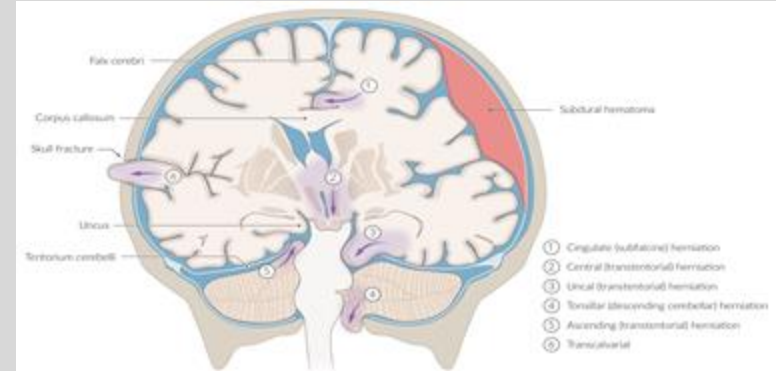
**Cushing Reflex** (Also known as “Cushing response”)

- A type of CNS ischemic response
- **Medical emergency – life-threatening!**

Possible causes: Head injury, intracranial hemorrhage, tumor, CNS inflammation/infection, excess CSF (hydrocephalus due to CSF overproduction or impaired CSF drainage), metabolic  $\Delta$  (ex: hyponatremia)

## CUSHING REFLEX TRIAD

- HYPERTENSION
- BRADYCARDIA
- RESPIRATORY DEPRESSION



# Atrial Reflex / BAINBRIDGE Reflex

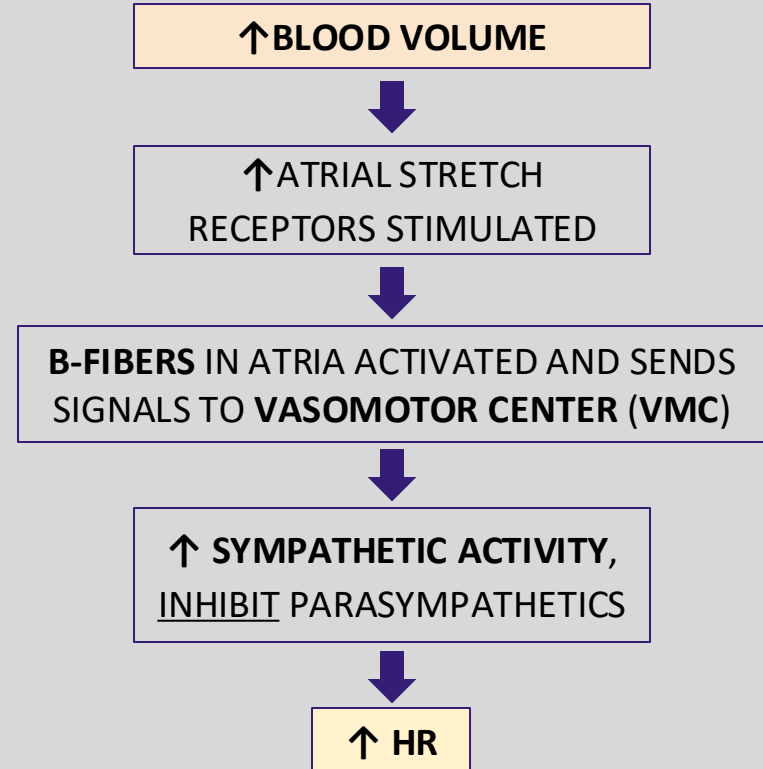
Physiologic reflex, mediated by **stretch receptors in the atria** (B-fibers)

## Role of Reflex:

- Maintain normal blood volume by *attempting* to eliminate excessive fluid in the circulation = **maintain cardiovascular system homeostasis.**

Increased venous return to the heart results in atrial distension, leading to  
↑HR

## MECHANISM



# Atrial Stretch and ANP

**Atrial stretch** causes a **non-neural** response from the stretching of the atrial cardiomyocytes → Secretion of **Atrial Natriuretic Peptide (ANP)**

- **ANP** acts via **cGMP**

## MECHANISM

↑BLOOD VOLUME



↑ATRIAL CARDIOMYOCYTES  
STRETCH



↑ANP SECRETION



- ↑RENAL EXCRETION OF NA<sup>+</sup> & WATER
- ↓NA<sup>+</sup> REABSORPTION AT RENAL COLLECTING TUBULE
- INHIBITION OF RENIN
- **DILATES AFFERENT** ARTERIOLES
- CONSTRICTS EFFERENT ARTERIOLES



## ANP Effects:

- Causes **vasodilation**
- ↓Na<sup>+</sup> reabsorption at renal collecting tubule
- **Dilates afferent renal arterioles & constricts efferent arterioles** → **Diuresis**

↑ RELAXATION OF VASCULAR SMOOTH MUSCLE CELLS

VASODILATION & ↓TPR

↓BP

DIURESIS



# **REVIEW QUESTIONS**

- Where are the baroreceptors located?
- Which cranial nerves are involved in baroreceptor reflex?
- Which part of the medulla receives afferent signals involved in baroreceptor reflex?
- Briefly describe baroreceptor reflex in the case of hypotension.
- Explain how  $\uparrow$ BP and  $\downarrow$ HR occur in Cushing reflex.

# REVIEW QUESTIONS - ANSWERS

**Where are the baroreceptors located?**

- Carotid sinus
- Aortic arch

**Which cranial nerves are involved in baroreceptor reflex?**

- CN IX (*Glossopharyngeal n.* in carotid sinus)
- CN X (*Vagus n.* in aortic arch)

**Which part of medulla receives afferent signals involved in baroreceptor reflex?**

- Nucleus tractus solitarius (NTS)

**Briefly describe baroreceptor reflex in the case of hypotension.**

[↓ BP] → [↓ Stretch detected by baroreceptors] → [↓ Baroreceptor firing rate] → [NTS receives signals and promotes sympathetic stimulation & decreases parasympathetic activity] ↓ → [↑ HR, ↑ SV, ↑ BP]

**Explain how ↑BP and ↓HR occur in Cushing reflex.**

[↑ ICP] → [Cerebral ischemia] → [↑ pCO<sub>2</sub> and ↓ pH] → [Compensatory activation of sympathetics] → [↑ BP] → [Baroreceptors detect ↑ BP] → [↑ Baroreceptor firing rate] → [↓ HR]



# PRACTICE QUESTION 1

During your 3rd year clerkship in Internal Medicine, you presented about blood pressure regulation.

A follow-up question was asked by a resident physician:  
*“How would the baroreceptor reflex be affected in patients who received vagotomy? Would it also have the same effect on the BP and HR?”*

# PRACTICE QUESTION 1 - ANSWER

If the vagus nerve (CN X) were damaged, baroreceptor input in the aortic arch would not be received by the NTS.

However, in contrast to baroreceptors in the carotid sinuses, the aortic arch baroreceptors are not very good at sensing low BP, thus, it would not make the brain think that there is hypotension.

Severing the vagus nerve would remove parasympathetic stimulation to the heart. Therefore, with the unopposed sympathetic stimulation in the heart, the HR would increase.

A 70-year-old man is brought to the emergency department due to progressive shortness of breath. He has had episodes of shortness of breath during minimal activities, like brushing his teeth. Medical history includes type 2 diabetes, hyperlipidemia, and a myocardial infarction 4 years ago. Oxygen saturation is 92% on room air. The patient presents with 3+ bilateral pitting edema in the legs. Which of the following is most likely increased in this patient?

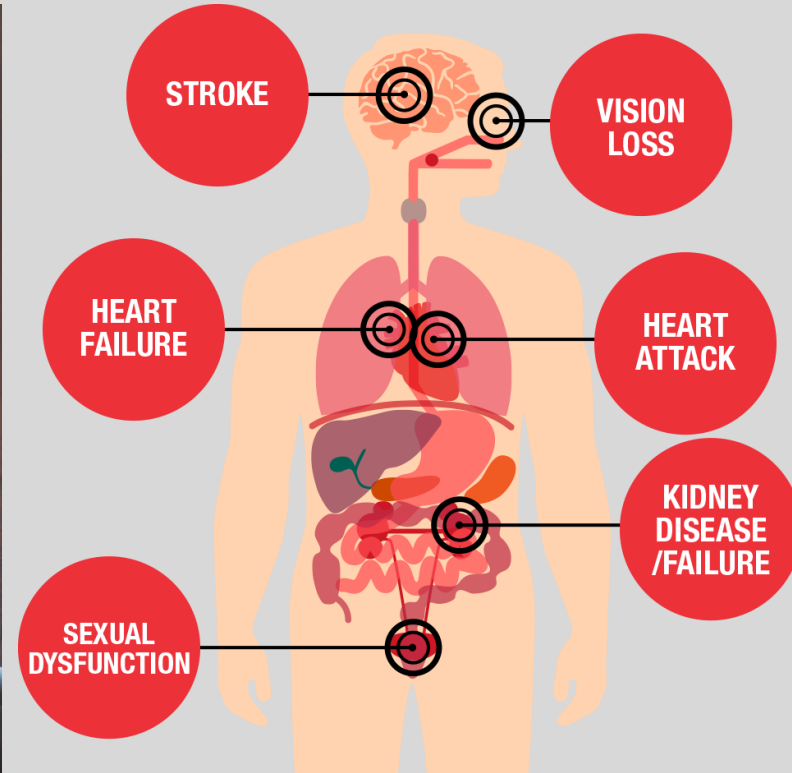
- a. Lung compliance
- b. Arteriolar resistance
- c. Renal perfusion
- d. Stroke volume



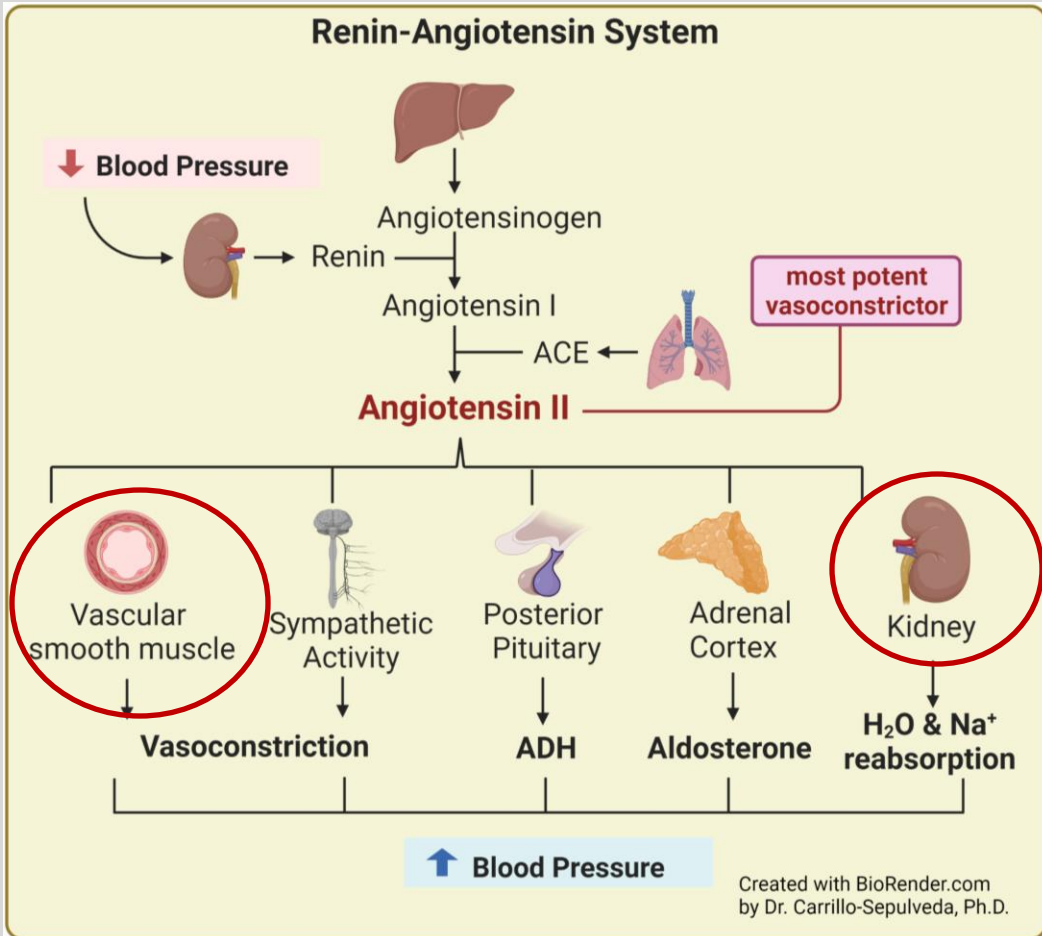
# LONG-TERM BP REGULATION

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# LONG-TERM BP REGULATION



- ♥ Regulate blood pressure within hours to days
- ♥ Occurs through hormonally-mediated pathways, primarily via **renin-angiotensin-aldosterone system (RAAS)**.

# RENAL BP REGULATION - JUXTAGLOMERULAR APPARATUS

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## MECHANISM

↓ Renal perfusion  
pressure (detected by  
baroreceptors in  
afferent arteriole)

↓ NaCl  
(hyponatremia)  
detected by  
macula densa

JG cells secrete renin

↑ Renin release

↑ Conversion of  
angiotensinogen to  
angiotensin I

↑ Conversion of  
angiotensin I to  
angiotensin II

afferent  
arteriole

Juxtaglomerular apparatus:  
macula densa  
juxtaglomerular or granular  
cells  
mesangial cells

efferent  
arteriole

nephron

Kidney

Created with BioRender by Dr. Carrillo-Sepulveda, Ph.D.

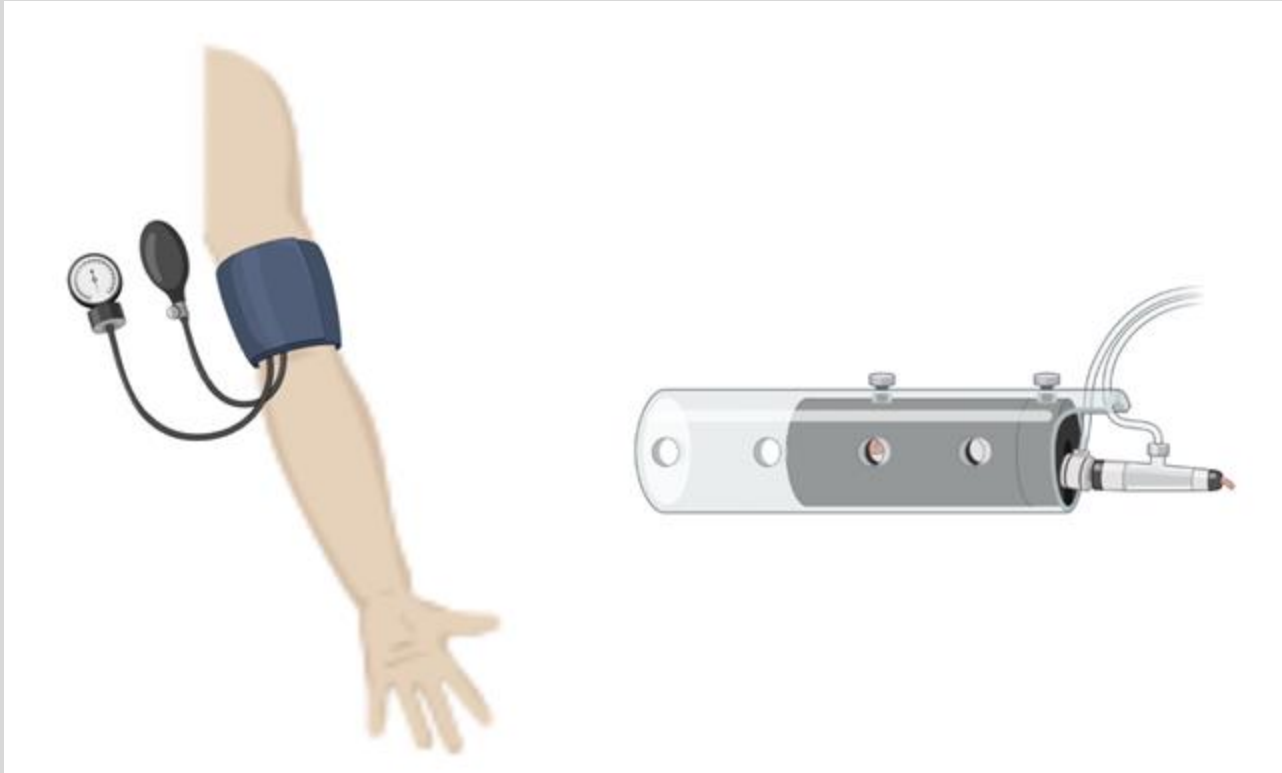
[Release of renin from the juxtaglomerular cells] → [RAAS activation] → [Direct vasoconstriction and ↑ extracellular volume through ↑ sodium and water reabsorption, ↓ K<sup>+</sup>, ↑ pH]

# INDIRECT BP MEASUREMENT

Blood pressure is usually measured indirectly

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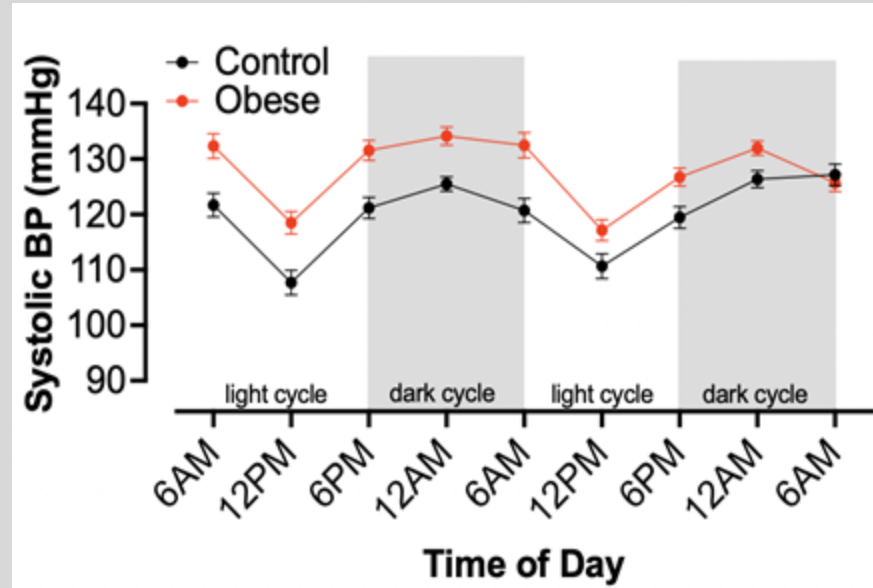
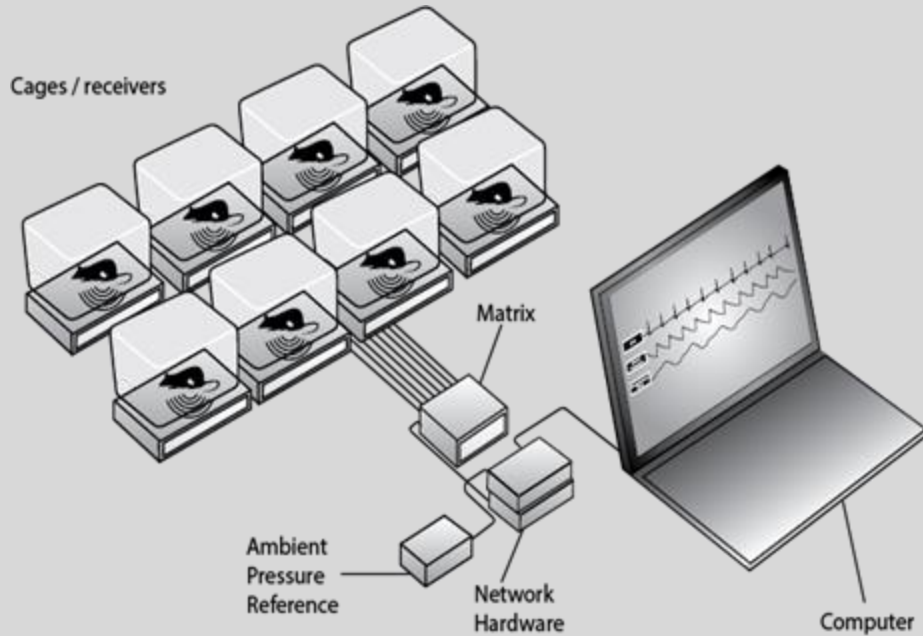


# TELEMETRY

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Research at Dr. Sepulveda's Lab



## Implantable radio telemetry

- Direct measurement of blood pressure
- Allows for continuous collection of BP data

# HTN: OTHER CONSIDERATIONS

## White Coat Hypertension

- Elevated BP in the clinical setting, but normal readings when BP measured elsewhere.

### Diagnosis:

In-Clinic:

BP  $\geq$  130/89 mmHg and  
 $\leq$  160/100 mmHg

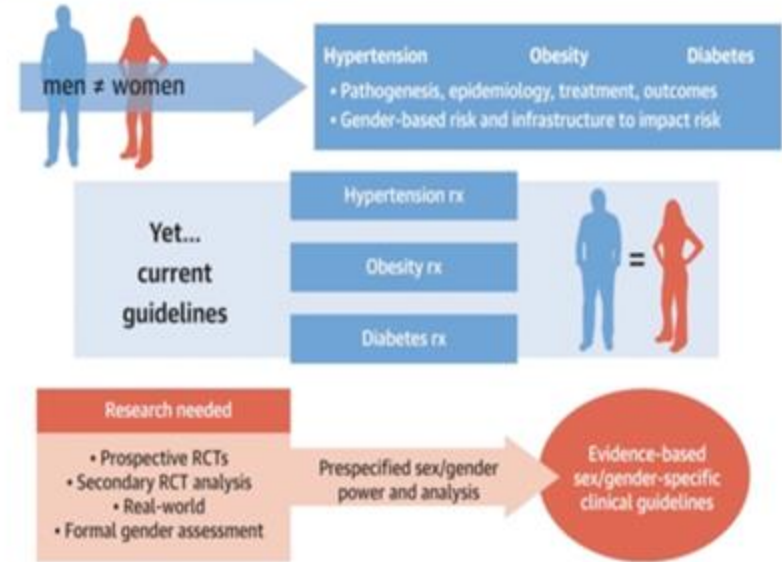
**BUT**

At Another Setting:

BP < 130/80 mmHg

## Sex Differences in BP Regulation

### CENTRAL ILLUSTRATION: A Call to Action for Studying Sex Differences



Regensteiner JG, et al. J Am Coll Cardiol. 2022;79(15):1492-1505.

# ACTIVATION OF SYSTEMIC RAAS

↓ MAP

↓ Renal perfusion pressure

Stretch receptors in JG cells  
sense ↓ renal perfusion  
pressure

↑ Renin release by granular\*  
cells into the bloodstream

RAAS activated

↑ Arteriolar vasoconstriction

$$\text{MAP} = \text{CO} \times \text{TPR}$$

\*Juxtaglomerular (JG) cells are also known as granular cells.

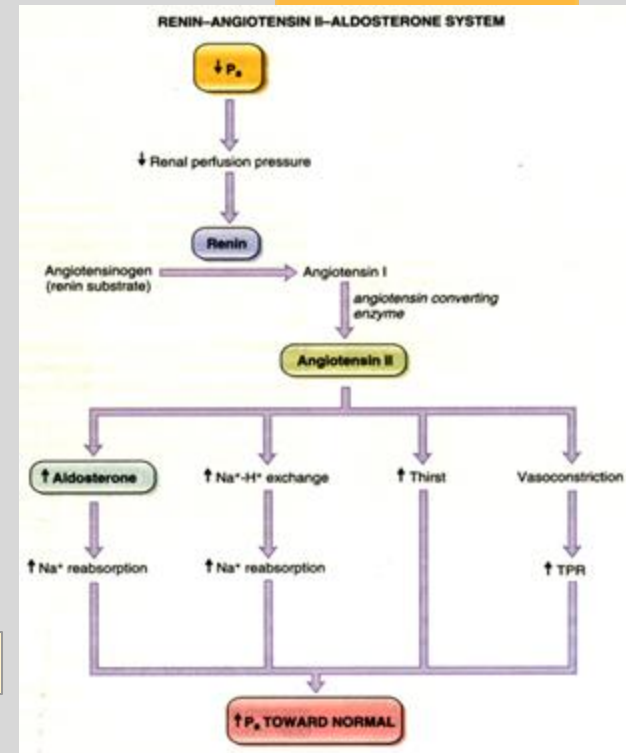
↑ Extracellular fluid & blood volume

↑ Venous  
return

↑ CO

↑ TPR

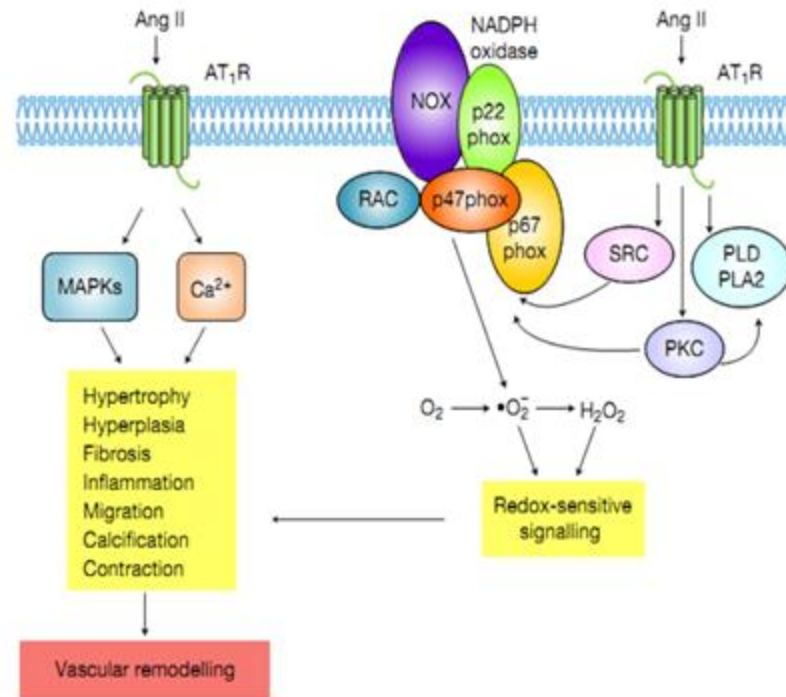
↑ MAP



# Angiotensin II- Other Effects

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Angiotensin-II-mediated signalling events in vascular smooth muscle cells that promote vascular remodelling and inflammation

expert reviews  
in molecular medicine  
Vol. 13; e11; March 2011

# CLINICAL RELEVANCE

## Hypertension (HTN)

### Risk Factors:

- ↑ Age
  - ♂ > ♀ Below age 65
  - ♀ > ♂ after menopause
- Obesity
- Diabetes
- High-sodium diet
- Excess EtOH
- Tobacco smoking
- Sedentarism
- Family history
- Higher rates in African American > White > Asian, Hispanic

| BLOOD PRESSURE CATEGORY                                      | SYSTOLIC mm Hg (upper number) | and/or | DIASTOLIC mm Hg (lower number) |
|--------------------------------------------------------------|-------------------------------|--------|--------------------------------|
| <b>NORMAL</b>                                                | LESS THAN 120                 | and    | LESS THAN 80                   |
| <b>ELEVATED</b>                                              | 120 – 129                     | and    | LESS THAN 80                   |
| <b>HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1</b>            | 130 – 139                     | or     | 80 – 89                        |
| <b>HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2</b>            | 140 OR HIGHER                 | or     | 90 OR HIGHER                   |
| <b>HYPERTENSIVE CRISIS (consult your doctor immediately)</b> | HIGHER THAN 180               | and/or | HIGHER THAN 120                |

American Heart Association, 2017

### Primary HTN:

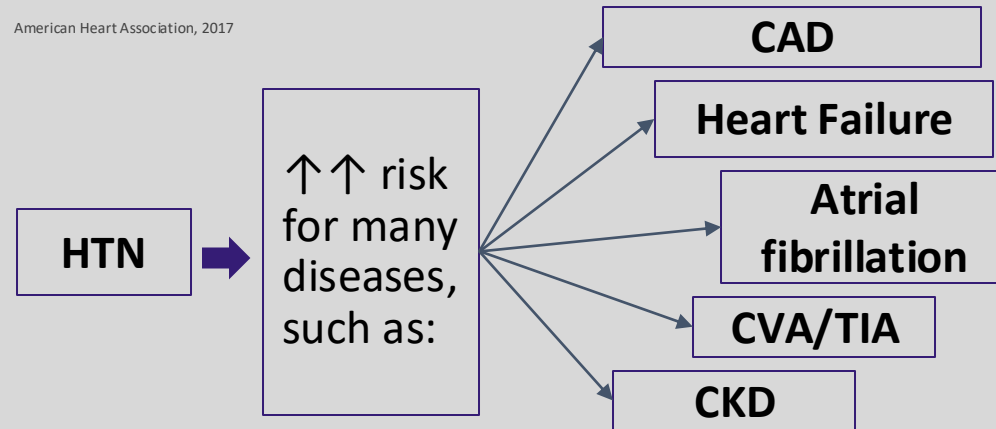
~90% of cases in adults, increasing prevalence in children and adolescents

- Related to ↑CO or ↑TPR

### Secondary HTN:

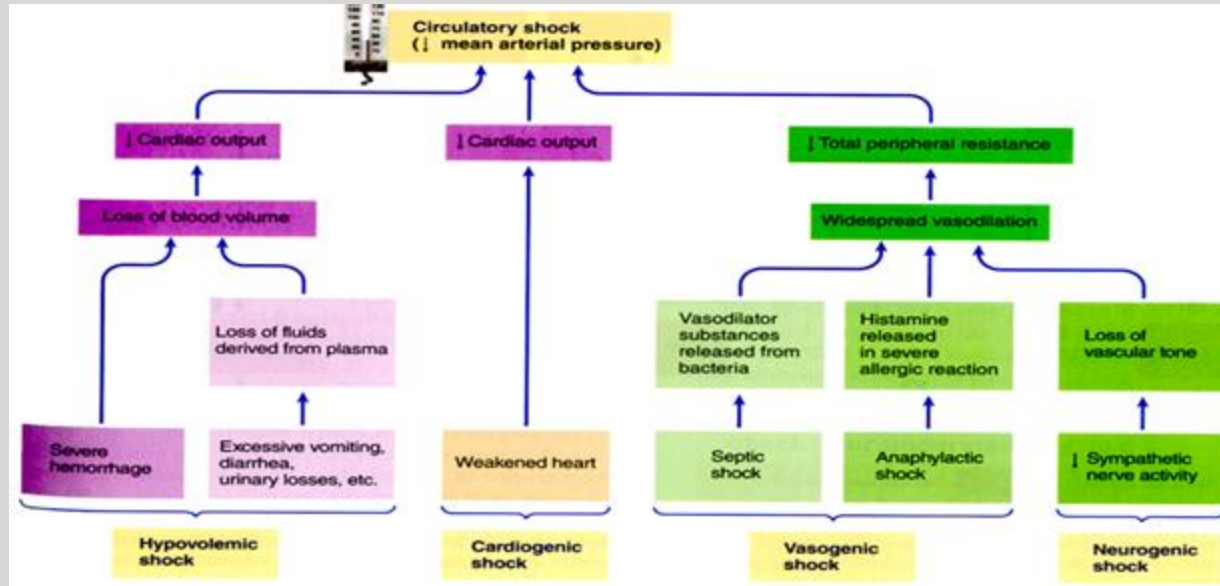
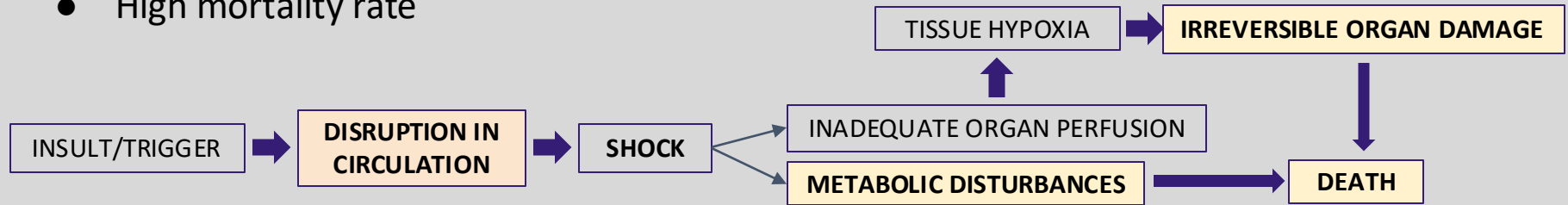
~10% of cases

- May be secondary to **renal or renovascular diseases**, such as fibromuscular dysplasia and atherosclerotic renal artery stenosis.

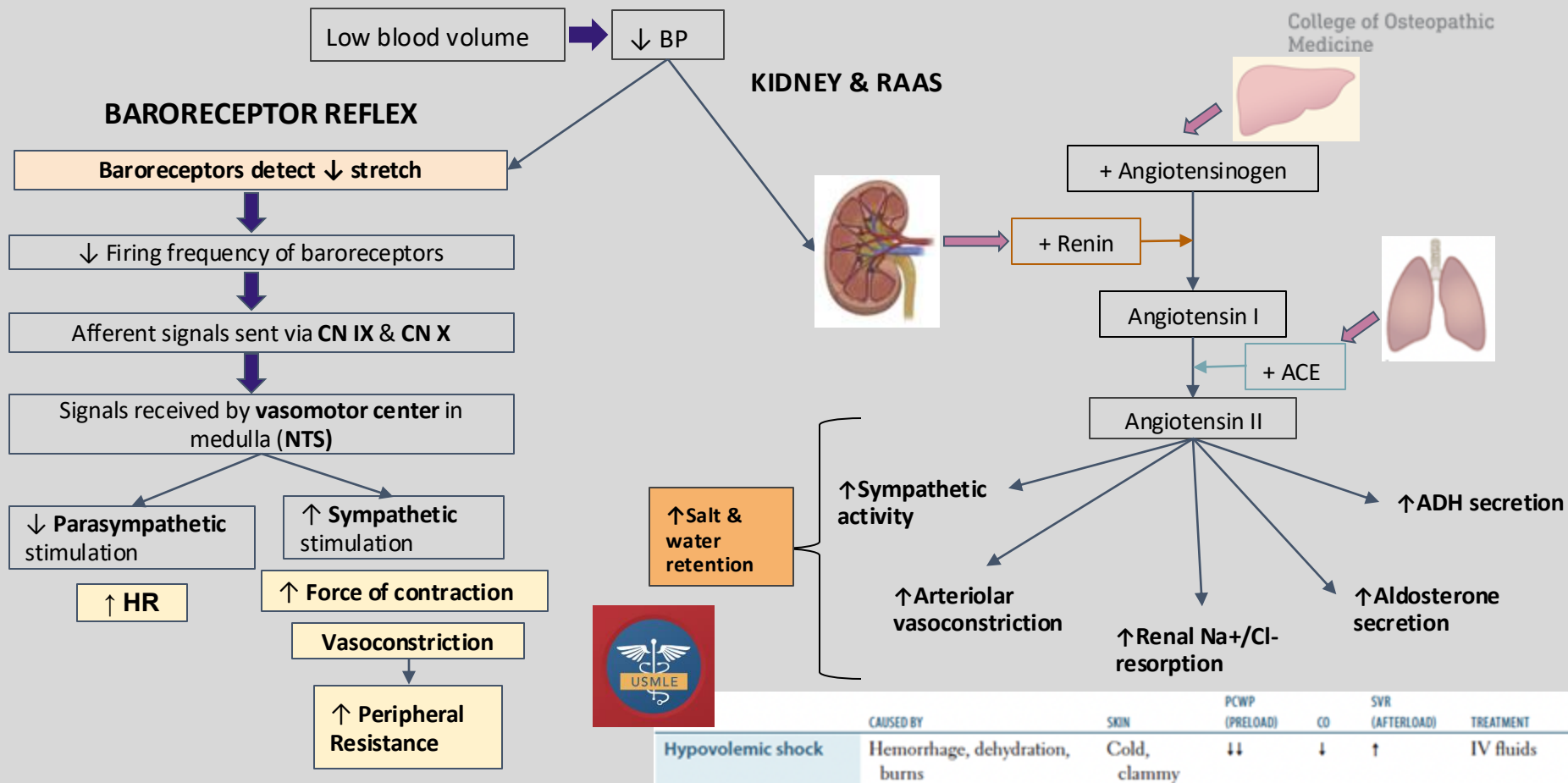


# SHOCK

- Life-threatening circulatory disorder
- High mortality rate



# HEMORRHAGE



# **REVIEW QUESTIONS**

- Where are the renal baroreceptors located?
- How does the kidney contribute to BP regulation and what is the net effect?
- Which type of renal cells secrete renin?
- Name 3 factors that trigger renin secretion.
- Briefly describe the role of renin in RAAS and BP regulation in response to hypotension.



# REVIEW QUESTIONS - ANSWERS

**Where are the renal baroreceptors located?**

- **Afferent arterioles**

**How does the kidney contribute to BP regulation and what is the net effect?**

- **Through secretion of renin, resulting in changes in salt and water retention**

**Which type of renal cells secrete renin?**

- **Juxtaglomerular (JG) cells, also known as granular cells**

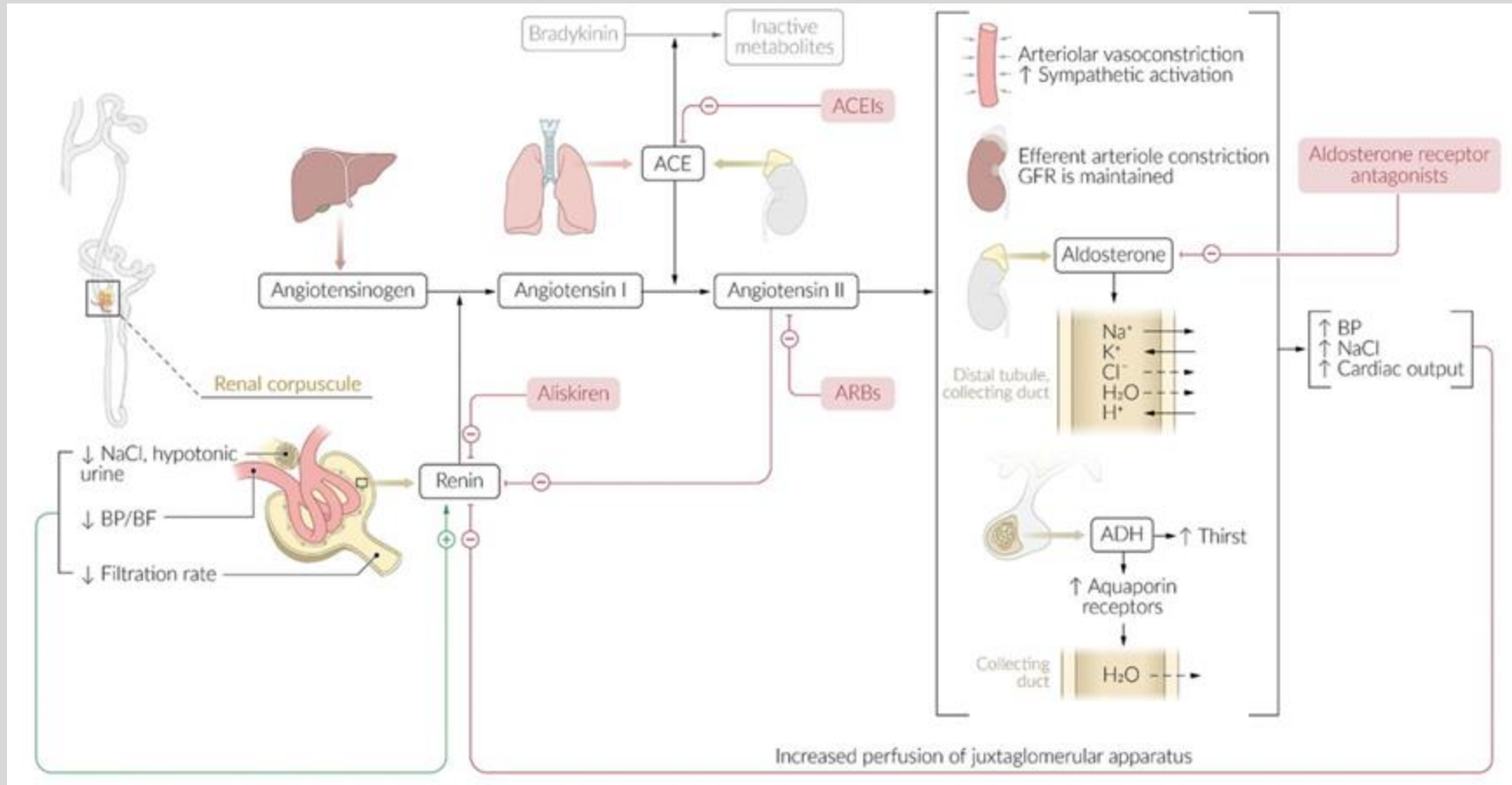
**Name 3 factors that trigger renin secretion.**

- **↓ Renal perfusion pressure (detected by baroreceptors in afferent arteriole)**
- **↑ Renal sympathetic discharge**
- **↓ NaCl delivery to macula densa cells.**

**Briefly describe the role of renin in RAAS and BP regulation in response to hypotension.**

**[↓ BP] → [↓ Renal perfusion pressure, ↓ NaCl delivery to macula densa cells] → [↑ Secretion of renin by JG cells] → [↑ Conversion of angiotensinogen to angiotensin I by renin] → [↑ Conversion of angiotensin I to angiotensin II by ACE] → [↑ Angiotensin II activity] → [↑ Sympathetic activity, arteriolar vasoconstriction, renal NaCl reabsorption, aldosterone secretion, ADH secretion] → [↑ Salt & water retention] → [↑ BP]**

# ANTIHYPERTENSIVES & RAAS



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# Thank you!

**QUESTIONS?**

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